FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.6: SURFACE AREA OF SOLIDS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

Your Final Explanation is your chance to answer the big driving question using everything you have learned across all 8 lessons. Think back to the hardware shop along Kirinyaga Road in Nairobi — the shopkeeper claimed that the frustum-shaped bucket uses the most sheet metal, even though all three buckets hold exactly 10 litres. Now you have the mathematical tools to test that claim yourself.

Write a well-organised explanation that answers BOTH parts of the driving question:

(1) How do we calculate exactly how much material is needed to cover or construct any solid shape?

(2) Why does the shape of a container matter — even when the volume stays the same?

Your explanation must include accurate surface area calculations for each of the three bucket shapes (cylinder, frustum, and cone), use correct mathematical vocabulary, and connect your working back to the real-world phenomenon. Use the section prompts below to structure your response. Read the exemplar answers carefully — they show the level of detail, reasoning, and mathematical precision you are aiming for.

Section 1: Setting Up the Problem — What Is Surface Area and Why Does It Matter?	
In your own words, explain what surface area means for a solid shape. Why is surface area the right measurement to use when a hardware shop owner in Nairobi wants to know how much sheet metal is needed to manufacture a bucket? Distinguish clearly between surface area and volume, and explain why two containers can have the same	Surface area is the total area of all the outer faces or curved surfaces of a solid shape — it is the measurement of how much flat material you would need if you unfolded (or 'unrolled') the entire outside of the shape and laid it flat. For the hardware shop owner on Kirinyaga Road, surface area is exactly the right measurement because sheet metal is a flat material that is cut and bent into shape; the more surface area a bucket has, the more sheet metal is required and the more expensive the bucket is to manufacture. Surface area and volume measure completely different things. Volume tells us how much space is inside a container — in this case, how many litres of water the bucket holds. Surface area tells us how much material wraps around the outside. It is entirely possible for two containers to have the same volume but very different surface areas, because the shape determines how the same 'space' is organised. A tall, narrow cylinder and a wide, flat cylinder can hold the same volume of water, yet their surface areas are different. The same principle applies to our three buckets: the cylindrical bucket, the frustum-shaped (tapered) bucket, and the conical bucket all hold exactly 10 litres, but their shapes cause their surface areas — and therefore the amount of sheet metal needed — to differ. This is precisely why the shopkeeper's claim deserves careful mathematical investigation.

volume but different surface areas. Use the three Kirinyaga Road buckets as your context.	

Section 2: Calculating the Surface Area of the Cylindrical Bucket	
Describe the net of a closed cylinder and identify each part. State the formula for the total surface area of a closed cylinder, explaining where each term comes from geometrically. Then apply the formula: assume the cylindrical bucket has a radius of 11.5 cm and a height of 24 cm. Show all your working clearly, give your answer to 2 decimal places, and state the units. (Use $\pi \approx 3.142$.)	When you unfold (net) a closed cylinder, you get three pieces: two identical circles (the top and bottom faces) and one rectangle (the curved lateral surface, which 'unrolls' into a flat sheet). The width of the rectangle equals the height of the cylinder, h, and its length equals the circumference of the circular base, $2\pi r$. Total Surface Area of a closed cylinder = 2 × (area of one circle) + area of the curved surface $= 2\pi r^2 + 2\pi rh$ $= 2\pi r(r + h)$ For the cylindrical bucket with $r = 11.5$ cm and $h = 24$ cm: Step 1 — Curved surface area: $2\pi rh = 2 \times 3.142 \times 11.5 \times 24$ $= 2 \times 3.142 \times 276$ $= 1{,}734.384$ cm² Step 2 — Area of the two circular faces: $2\pi r^2 = 2 \times 3.142 \times (11.5)^2$ $= 2 \times 3.142 \times 132.25$ $= 831.299$ cm² Step 3 — Total surface area: $1{,}734.384 + 831.299 = 2{,}565.68$ cm² (to 2 d.p.) This means that to manufacture one closed cylindrical bucket of this size, the factory would need approximately 2,565.68 cm² of sheet metal. Note: if the bucket is open at the top (no lid), you subtract one circle: $2{,}565.68 - 415.65 = 2{,}150.03$ cm².

Section 3: Calculating the Surface Area of the Conical Bucket	
Describe the net of a cone and identify each part. State the formula for the total surface area of a closed cone, explaining what the slant height is and how it differs from the	When you unfold a closed cone, you get two pieces: one circle (the base) and one sector of a larger circle (the curved lateral surface). The radius of that sector is the slant height, l, of the cone — the distance measured along the sloping side from the apex (tip) down to the edge of the base. The slant height is always longer than the vertical height, h, because it is the hypotenuse of the right-angled triangle formed by h and the base radius r.

vertical height. Then apply the formula: assume the conical bucket has a base radius of 14 cm and a vertical height of 20 cm. Show all your working, including how you calculate the slant height using Pythagoras' theorem. Give your answer to 2 decimal places. (Use $\pi \approx 3.142$.)	Finding the slant height using Pythagoras' theorem: $l^2 = r^2 + h^2$ $l^2 = 14^2 + 20^2$ $l^2 = 196 + 400 = 596$ $l = \sqrt{596} \approx 24.41$ cm Total Surface Area of a closed cone = area of base + area of curved surface $= \pi r^2 + \pi rl$ $= \pi r(r + l)$ Step 1 — Area of the base: $\pi r^2 = 3.142 \times 14^2$ $= 3.142 \times 196$ $= 615.83$ cm² Step 2 — Curved (lateral) surface area: $\pi rl = 3.142 \times 14 \times 24.41$ $= 3.142 \times 341.74$ $= 1{,}073.79$ cm² Step 3 — Total surface area: $615.83 + 1{,}073.79 = 1{,}689.62$ cm² (to 2 d.p.) The cone uses considerably less sheet metal than the cylinder we calculated in Section 2. This already suggests that the three bucket shapes are NOT equally expensive to produce, even though they all hold 10 litres.

Section 4: Calculating the Surface Area of the Frustum-Shaped Bucket	
A frustum is what remains when the top of a cone is cut off with a cut parallel to the base. Describe the net of a frustum and identify each face. State the formula for the total surface area of a frustum, defining every variable. Then apply the formula: assume the frustum bucket has a larger base radius R = 14 cm, smaller top radius r = 9 cm, and vertical height h = 22 cm. Show all working,	A frustum is a cone with its pointed top removed by a horizontal cut. When you unfold a frustum, you get three pieces: one large circle (the base, radius R), one smaller circle (the top opening, radius r), and one curved band — which is actually the difference between two cone sectors — that forms the lateral surface. Finding the slant height of the frustum: The slant height l is the distance along the sloping side between the two circular edges. $l = \sqrt{h^2 + (R - r)^2}$ $l = \sqrt{22^2 + (14 - 9)^2}$ $l = \sqrt{484 + 25}$ $l = \sqrt{509}$ $l \approx 22.56$ cm

including how you calculate the slant height of the frustum. Give your answer to 2 decimal places. (Use $\pi \approx 3.142$.) Finally, compare all three results and evaluate the shopkeeper's claim.	Total Surface Area of a frustum = area of base + area of top + lateral surface area $= \pi R^2 + \pi r^2 + \pi(R + r)l$ Step 1 — Area of large base: $\pi R^2 = 3.142 \times 14^2 = 3.142 \times 196 = 615.83$ cm² Step 2 — Area of small top: $\pi r^2 = 3.142 \times 9^2 = 3.142 \times 81 = 254.50$ cm² Step 3 — Lateral surface area: $\pi(R + r)l = 3.142 \times (14 + 9) \times 22.56$ $= 3.142 \times 23 \times 22.56$ $= 3.142 \times 518.88$ $= 1{,}630.29$ cm² Step 4 — Total surface area: $615.83 + 254.50 + 1{,}630.29 = 2{,}500.62$ cm² (to 2 d.p.) --- Comparison of All Three Buckets (same 10-litre volume) --- – Cylindrical bucket: ≈ 2,565.68 cm² – Frustum-shaped bucket: ≈ 2,500.62 cm² – Conical bucket: ≈ 1,689.62 cm² Evaluating the shopkeeper's claim: The shopkeeper said the frustum uses the most metal. Based on these calculations, the frustum (≈ 2,500.62 cm²) actually uses slightly LESS sheet metal than the cylinder (≈ 2,565.68 cm²) for these particular dimensions, while the cone uses far less than either. The shopkeeper's claim is NOT necessarily correct — it depends on the exact dimensions. What is true, and what the shopkeeper may have been observing, is that the frustum's net is more complex than the others, which can make it more expensive to cut and assemble in the factory, even if the raw material area is similar to the cylinder. This shows that surface area alone does not always determine manufacturing cost, but it is the essential starting point for the calculation.

Section 5: Answering the Driving Question — Shape, Surface Area, and Real-World Decisions	
Now bring everything together. In a well-written paragraph (or series of paragraphs), answer BOTH parts of the driving question fully: **(1) How do we calculate exactly**	To calculate exactly how much material is needed to cover or construct any solid shape, you must find its total surface area — the sum of the areas of every face and curved surface that makes up the outside of the solid. For each type of solid, mathematicians have derived a specific formula based on the geometry of its faces. For a cylinder, you add two circular faces and a rectangular curved surface (TSA = $2\pi r^2 + 2\pi rh$). For a cone, you add a circular base and a curved sector (TSA = $\pi r^2 + \pi rl$, where l is the slant height found using Pythagoras' theorem). For a frustum — the trickiest of the three — you add two circles of different sizes and a curved lateral band (TSA = $\pi R^2 + \pi r^2 + \pi(R + r)l$). In every case, the key steps are the same: identify all faces from the net of the

how much material is needed to cover or construct any solid shape? **(2) Why does the shape of a container matter, even when the volume stays the same?** **Reference your calculations, the Kirinyaga Road buckets, and at least ONE other real-life Kenyan context where surface area matters (e.g., roofing iron sheets in the Rift Valley, packaging for mandazi or maize flour, water tanks in Kisumu, paint for grain silos). Reflect on what surprised you or what you would now advise the shopkeeper.**	solid, recall or derive the correct formula, substitute the given dimensions carefully, and calculate step by step. Shape matters enormously even when volume stays the same, because volume and surface area are governed by different mathematical relationships. A tall narrow shape 'packs' the same volume using a different arrangement of surface than a wide flat shape. Our three 10-litre buckets proved this exactly: the cylindrical bucket required about 2,565.68 cm² of sheet metal, the frustum about 2,500.62 cm², and the cone only about 1,689.62 cm² — even though all three hold identical amounts of water. The cone achieves the same storage capacity with far less material because its shape tapers sharply, reducing the surface area dramatically. This principle appears across many real-life situations in Kenya. Families in the Rift Valley who are roofing a new home need to calculate the surface area of their roof accurately before buying iron sheets — buying too little means a second trip to the hardware store, and buying too much wastes money. Factories in Nairobi that package maize flour or mandazi in cylindrical or box-shaped containers use surface area calculations to minimise the amount of packaging material used, cutting costs. Water engineers in Kisumu designing cylindrical storage tanks calculate surface area to estimate the amount of metal or concrete needed for construction. If I were to advise the shopkeeper on Kirinyaga Road, I would say: your intuition that the frustum is more expensive is reasonable, but the mathematical reason is more nuanced than simply 'it uses the most metal.' For these particular bucket dimensions, the frustum and cylinder use similar amounts of sheet metal, while the cone uses significantly less. The frustum may cost more because its curved, tapered shape is harder to cut and weld accurately in the factory, increasing labour costs even if the raw material area is comparable. The most important lesson from this unit is that you cannot judge material usage by volume alone — you must always calculate the surface area using the correct formula for the specific shape.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding of Surface Area vs. Volume	Clearly and accurately distinguishes surface area from volume using precise mathematical language. Consistently connects surface area to material usage in the Kirinyaga Road context and at least one other Kenyan context. Explains convincingly why equal volumes do not imply equal surface areas.	Correctly distinguishes surface area from volume and links surface area to material usage. Makes some connection to the Kirinyaga Road phenomenon. Explanation is mostly clear with minor imprecision in language.	Shows partial understanding; may confuse surface area and volume at points, or explain one correctly but not the other. Connection to the bucket phenomenon is weak or missing. Mathematical language is imprecise.
Accuracy of Surface Area Calculations (Cylinder, Cone, Frustum)	All three surface area calculations are fully correct, with all steps shown clearly. Slant heights are correctly derived using Pythagoras' theorem where required. Final answers are correctly rounded to 2 decimal places with correct units (cm²). Working is logically sequenced and easy to follow.	At least two of the three calculations are correct with most steps shown. Minor arithmetic errors may be present but the method is sound. Slant heights are mostly correct. Units are stated.	Only one calculation is attempted correctly, or significant errors in method and arithmetic are present across calculations. Slant height may be missing or incorrect. Steps are incomplete or difficult to follow.
Correct Use and Explanation of	States the correct formula for each solid	States the correct formula for each solid	One or more formulae are incorrect or

Formulae	(cylinder, cone, frustum) and explains what each variable represents geometrically (e.g., why 2πrh is the lateral surface of a cylinder; why π(R+r)l is the lateral surface of a frustum). Formula for slant height is stated and applied correctly.	and uses it correctly in calculations. Geometric explanation of each term is partial but mostly accurate. Slant height formula is used correctly.	missing. Limited or no geometric explanation of formula terms. Slant height may be confused with vertical height.
Evaluation of the Shopkeeper's Claim Using Evidence	Directly compares the three calculated surface areas and uses the numerical evidence to evaluate the shopkeeper's claim with precision. Correctly identifies which bucket uses the most material based on their calculations and acknowledges that the answer depends on dimensions. Recognises that manufacturing cost may differ from material cost alone.	Compares the three surface area results and gives a clear verdict on the shopkeeper's claim. The evaluation is based on the calculated values. May not fully explore why manufacturing complexity could affect cost beyond material area.	Makes a judgement about the shopkeeper's claim but does not clearly link it to the calculated surface area values. Comparison of the three shapes is vague or based on intuition rather than mathematical evidence.
Real-World Connection and Communication	Integrates at least two Kenyan real-world contexts (including the Kirinyaga Road phenomenon) naturally and accurately throughout the explanation. Writing is clear, well-organised, and uses correct mathematical vocabulary throughout. The explanation would genuinely help another student understand both the mathematics and its relevance.	References the Kirinyaga Road context clearly and includes at least one other Kenyan real-world context. Writing is organised and mostly uses correct vocabulary. A peer could follow the explanation with minor difficulty.	Mentions the Kirinyaga Road buckets but does not develop real-world connections meaningfully. Writing is unclear or poorly organised in places. Mathematical vocabulary is limited or incorrectly used.